

VERSION CONTROL REPORT

1. WHAT IS VERSION CONTROL?

The system referred to as version control monitors changes on a file or set files over time and enables you to track their history and revert to previous versions later. It documents all such changes made by other developers and maintains a historical record of the changes to the program in a particular database. That allows you to compare changes over time, restore the entire project back to a previous state, restore individual files to a previous state, and see who last made a modification that could possibly be the source of a problem.

2. WHY IS VERSION CONTROL IMPORTANT

It is a safety net for code integrity: it allows the developer to see each commit and access, review, collaborate, experiment, compare, and undo changes.

- **Collaboration:** This allows different developers to work in the same project at the same time. The Version control systems enable teams to avoid conflicts and encourage chances of experimentation and innovation by keeping track of all the changes made.
- **Reduce error:** A version control tool tracks every variation of the file; it allows engineers to debug fast and roll back the project. It acts as a safety net because it prevents accidental deletion of code.
- **Branching and merging:** A version control tool tracks every variation of the file; it allows engineers to debug fast and roll back the project. It acts as a safety net because it prevents accidental deletion of code.

3. TYPES OF VERSION CONTROL

- **Local version control systems (LVCS):** On the developer's computer, changes may be tracked locally. Modifications to files are kept within patches. Still, this method works only for very small projects and does not support the collaboration necessary for some types of projects
- **Centralized version control systems (CVCS):** All of the files maintain their presence on one central server. Changes can be made by users by checking out files from the server and then committing back into the central repository. Every user commits to the same branch and works with the same repository.
- **Distributed version control systems (DVCS):** The repository is available to the user anywhere. Contributors can even work on the same codebases if they are not on the same network; however, collaboration is encouraged. In this way, without depending on anyone, the user can commit, branch, or merge locally.

4. WHAT IS GITHUB?

GitHub is a web-based platform that allows developers to collaborate on code, share their work and manage projects. It is a based-on Git, an open-source distributed version control system, to track changes to code and recall specific versions later. GitHub hosts Git repositories, enabling users to clone, push, pull and collaborate on them.

5. IMPORTANT GIT COMMANDS

- **Git commit:** It saves a version of your project to the repository. Along with that, one can also add a message that describes the changes.
- **Git Branch:** This will give a list of all the branches he has in the Git repository. A new branch can also be created by using it.
- **Git checkout:** Allows switching between branches or revert files back to a previous state. It enables the creation of a new branch and modifications you can make on it and work on.
- **Git show:** Displays modifications done on a branch. It shows changes and commits against a codebase.
- **Git merge:** Links changes from one branch to another.
- **Git pull:** To fetch and merge changes from the remote repository to your local repository. It is a combination of both git fetch and git merge.
- **Git push:** To upload the changes made to a remote repository
- **Git status:** It is a command to show the working directory status. In addition, it gives the changes that are made

CONCLUSION

This is an important tool for the modern management of documents and software because it facilitates smooth communication, traceable change, and preservation of a history of work. Version control, today, includes industry standards for using Git and GitHub, in which all developers worldwide would like to use this feature-rich, distributed architecture tool.

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